

PROJECT DOCUMENTATION: LAUNCH YOUR LINUX MACHINE ON AWS WITH PUTTY



▪ PROJECT OVERVIEW :

In this project we are show the how to create a **Linux server on AWS EC2** and connect to it securely using **PuTTY**. It includes creating an EC2 instance, using a key file, and accessing the server through SSH. This project helps us learn the basics of **AWS, Linux, and Cloud Computing** 🚀

▪ REQUIREMENT 🔑 :

Before launching your Linux machine on AWS and connecting via PuTTY, ensure you have the following requirement ready:

- **AWS Account** : An active Amazon Web Services account with permissions to launch EC2 instances.
- **PuTTY Installed** : The PuTTY desktop application installed on your local machine (includes both PuTTY and PuTTYgen).
- **Network Access** 🌐 : A stable internet connection to access the AWS Management Console and establish the SSH connection.
- **Key Pair** 🔑 : Access to download the .pem private key file generated by AWS during instance setup.
- **Security Group Rules** : Awareness of your local public IP address to restrict **Port 22** access to your machine only (recommended for security).

- **WHAT IS LINUX?** 🌐

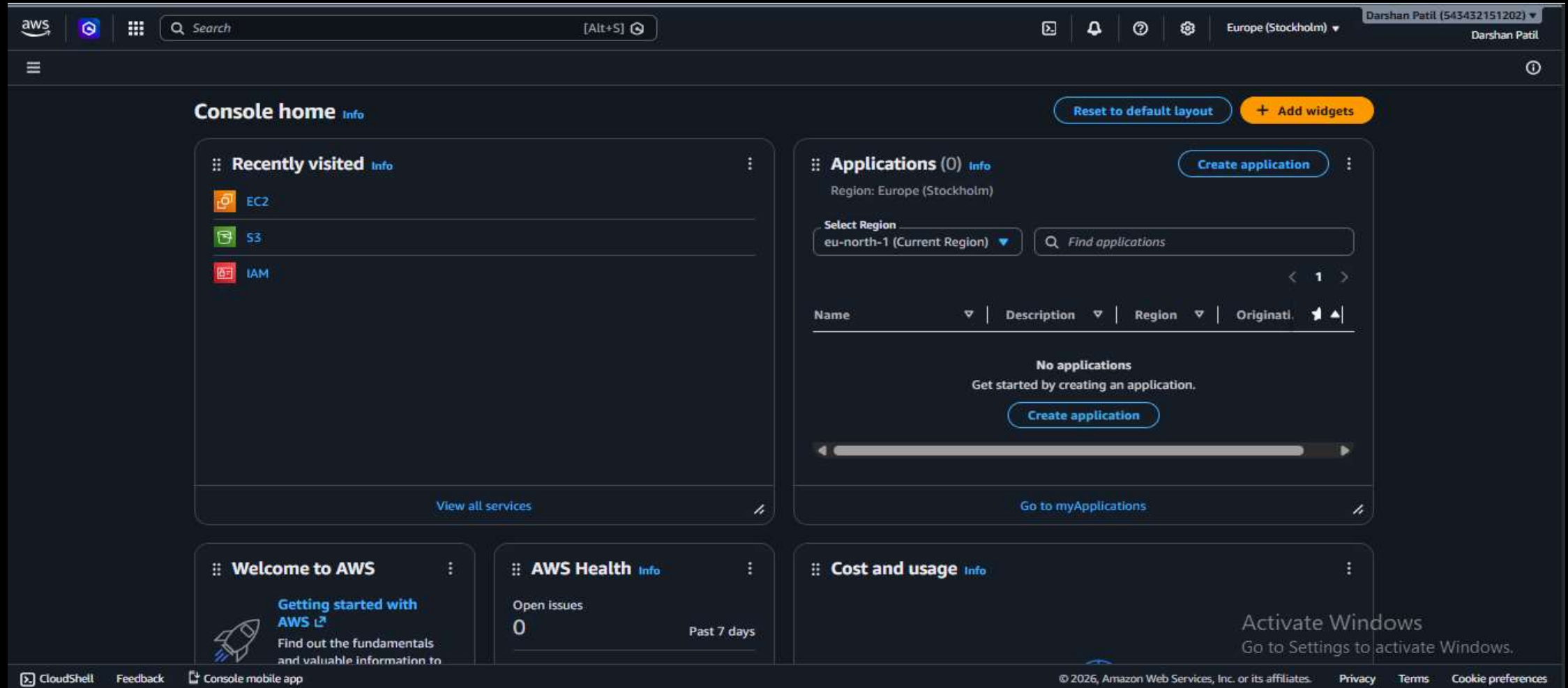
Linux is a free and open-source operating system used to manage computers and servers. It is popular because it is secure, fast, and reliable.

- **WHAT IS PUTTY?** 💻

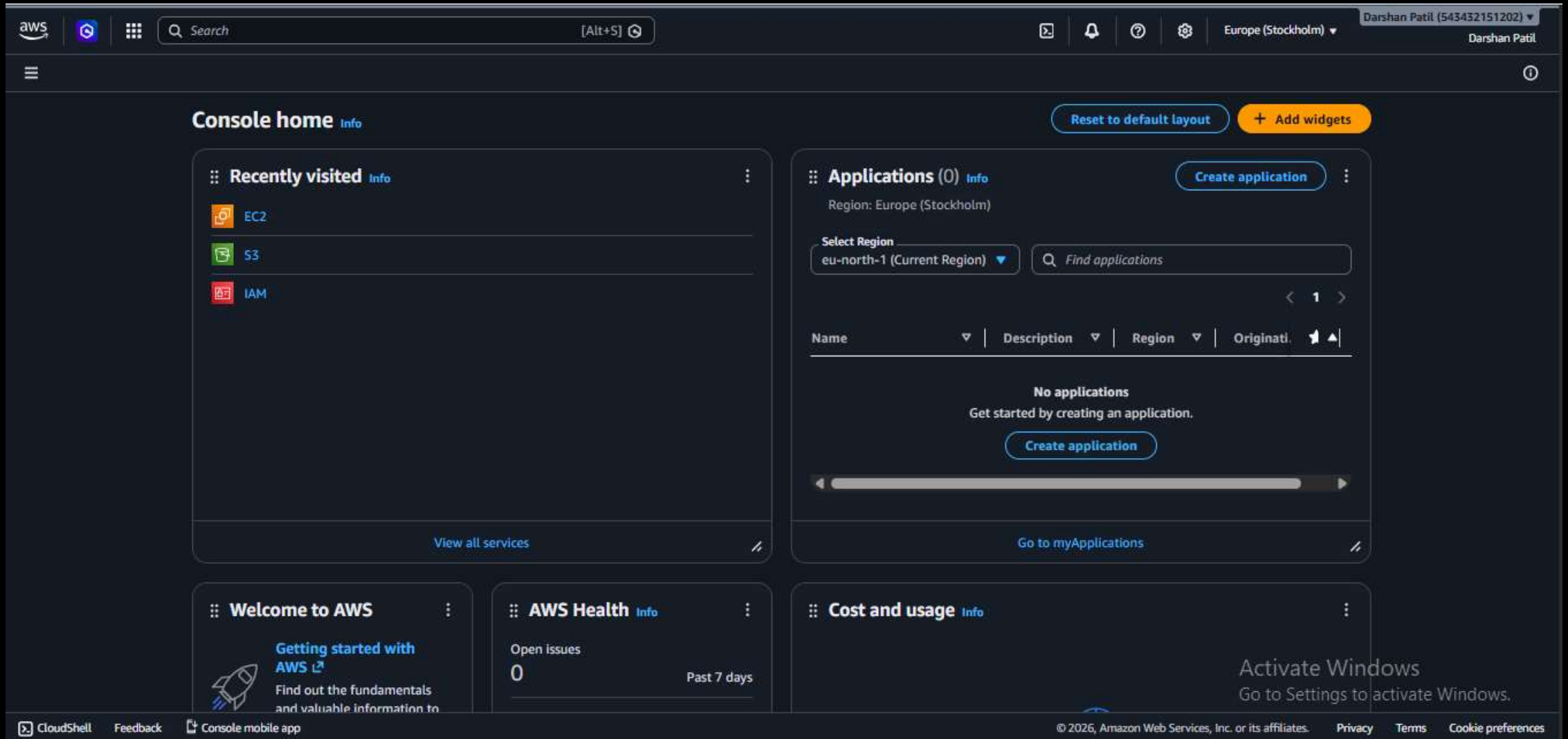
PuTTY is a free and open-source SSH client that allows you to securely connect to remote servers, typically Linux-based, from a Windows machine. It enables you to access and manage your server via the command line. 💻 🔑

I. LAUNCH YOUR EC2 INSTANCE ON AWS 🚀

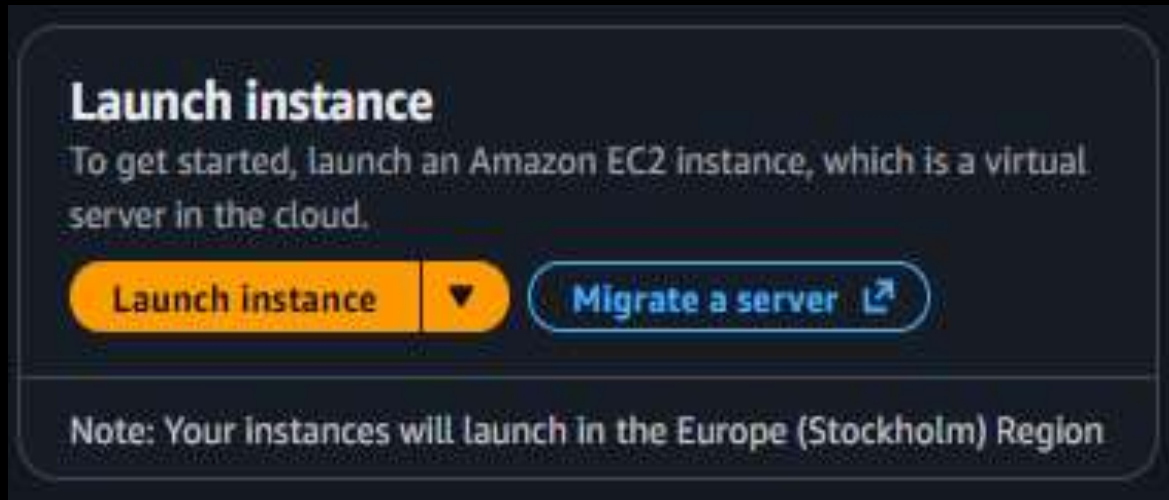
1. Log in to your **AWS Management Console**. It's time to get started.



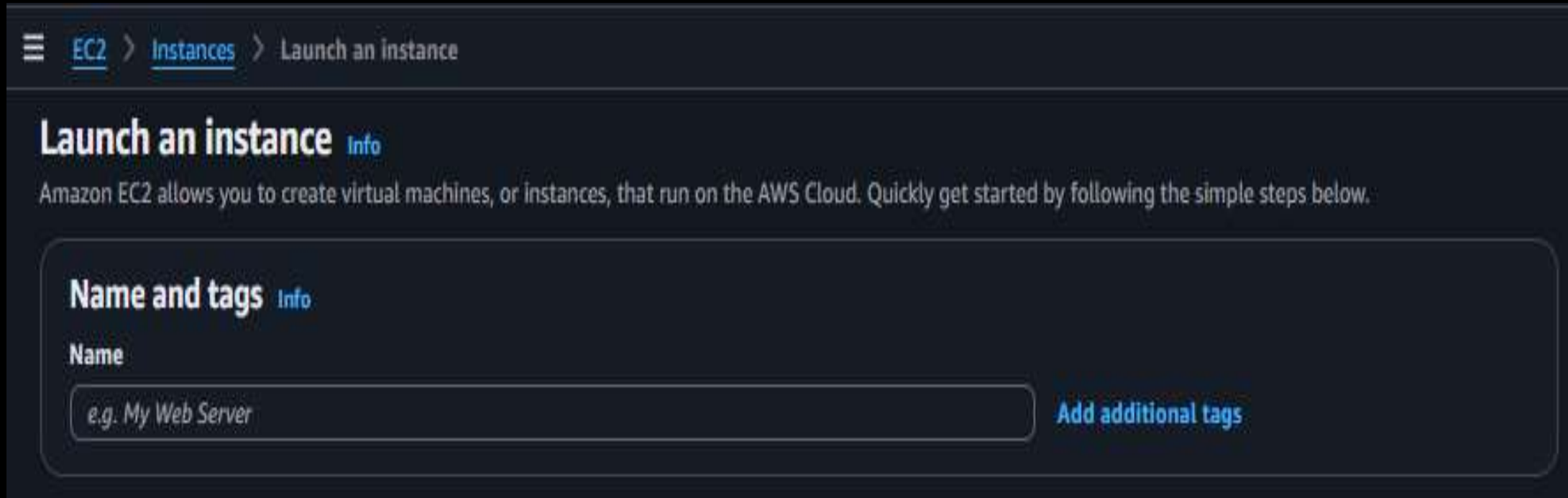
2. Head over to **EC2 Dashboard** under **Services** , the heart of your cloud adventure



3. Click the **Launch Instance** button and get ready to takeoff ✈️



4. Give a name to your instance , this helps you easily identify it later. 💡
(Eg – creating-linux-using-putty)



The screenshot shows the AWS Management Console interface for launching an EC2 instance. At the top, there is a breadcrumb navigation path: 'EC2 > Instances > Launch an instance'. Below this, the main heading is 'Launch an instance' with an 'Info' link. A descriptive text states: 'Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.' The first step is 'Name and tags', also with an 'Info' link. Under this step, there is a 'Name' label and a text input field containing the placeholder text 'e.g. My Web Server'. To the right of the input field is a button labeled 'Add additional tags'.

5. Pick your Amazon Machine Image (AMI) (e.g... Ubuntu or Amazon Linux)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux



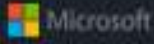
macOS



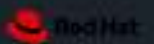
Ubuntu



Windows



Red Hat



SUSE Linux



Debian





[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0402e980e69d5978b (64-bit (x86), uefi-preferred) / ami-033fd902867ca7065 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

6. **Choose an Instance Type** : Select **t2.micro** , which is perfect for beginner and falls within free tier offering 🎁.

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.micro **Free tier eligible**

Family: t2 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

7. In the **Key Pair** section, click on **Create a new pair** to generate a new key pair for secure access 🔑

▼ **Key pair (login)** [Info](#)

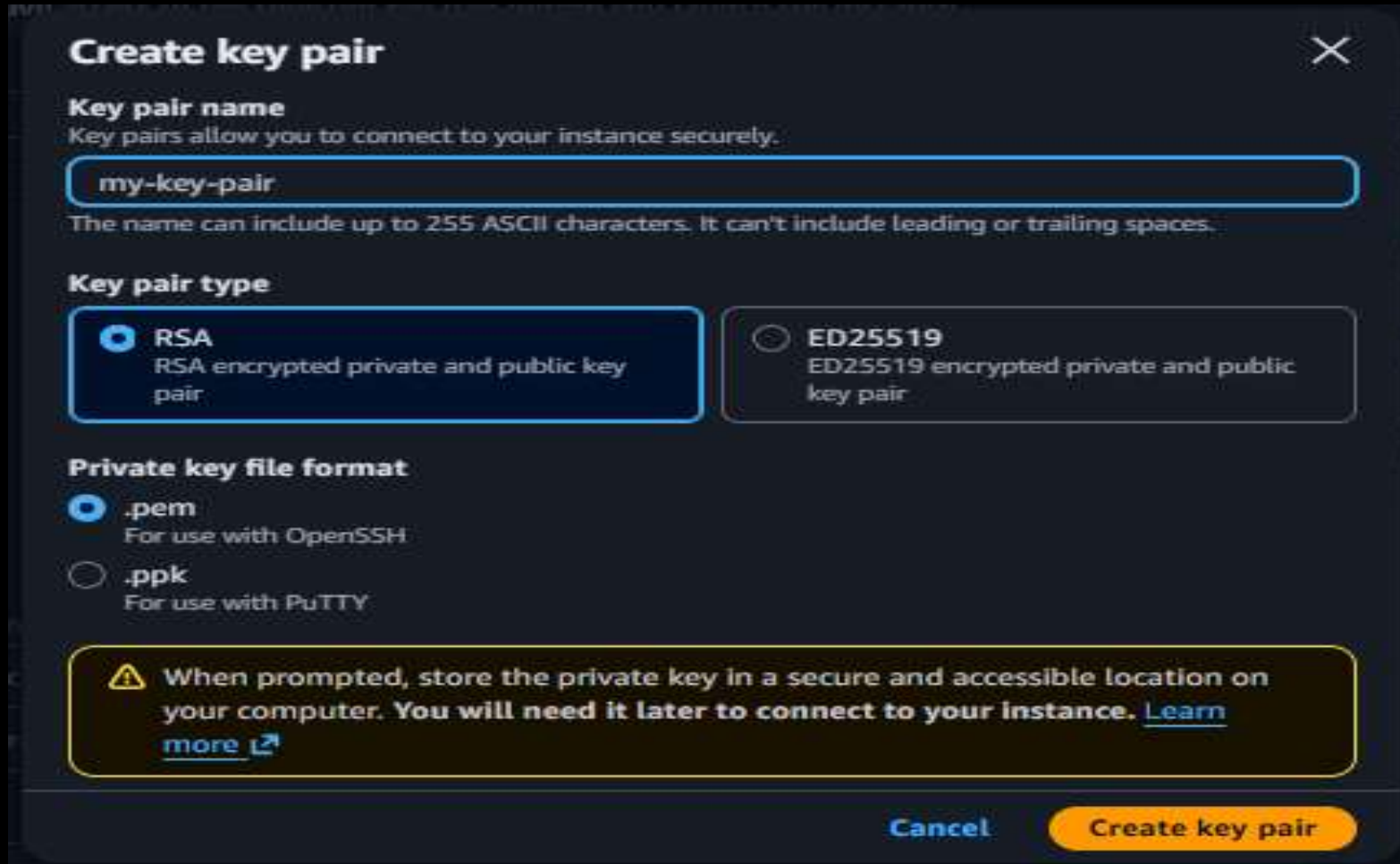
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select ▼

[Create new key pair](#)

8. Give your key pair a name (e.g.. **my-key-pair**) 📄



The screenshot shows the 'Create key pair' dialog box in AWS. It has a title bar with a close button (X). The main content is divided into sections: 'Key pair name' with a text input field containing 'my-key-pair' and a note about character limits; 'Key pair type' with two radio button options, 'RSA' (selected) and 'ED25519'; and 'Private key file format' with two radio button options, '.pem' (selected) and '.ppk'. At the bottom, there is a warning box with a yellow triangle icon and a 'Create key pair' button.

Create key pair ✕

Key pair name
Key pairs allow you to connect to your instance securely.

my-key-pair

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

Warning: When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel **Create key pair**

9. Configure your **Security Group** , just make sure **SSH** (port 22) is allowed 🔑.

▼ **Network settings** [Info](#)

Network [Info](#)

vpc-0ca3dd3c304fb2fcd

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠️ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

10. See the instance **Summary** and click on **Launch Instance** 🖨

▼ **Summary**

Number of instances | Info

1

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

Cancel

Launch instance

🔗 Preview code

What is PuTTYgen ? 🔑

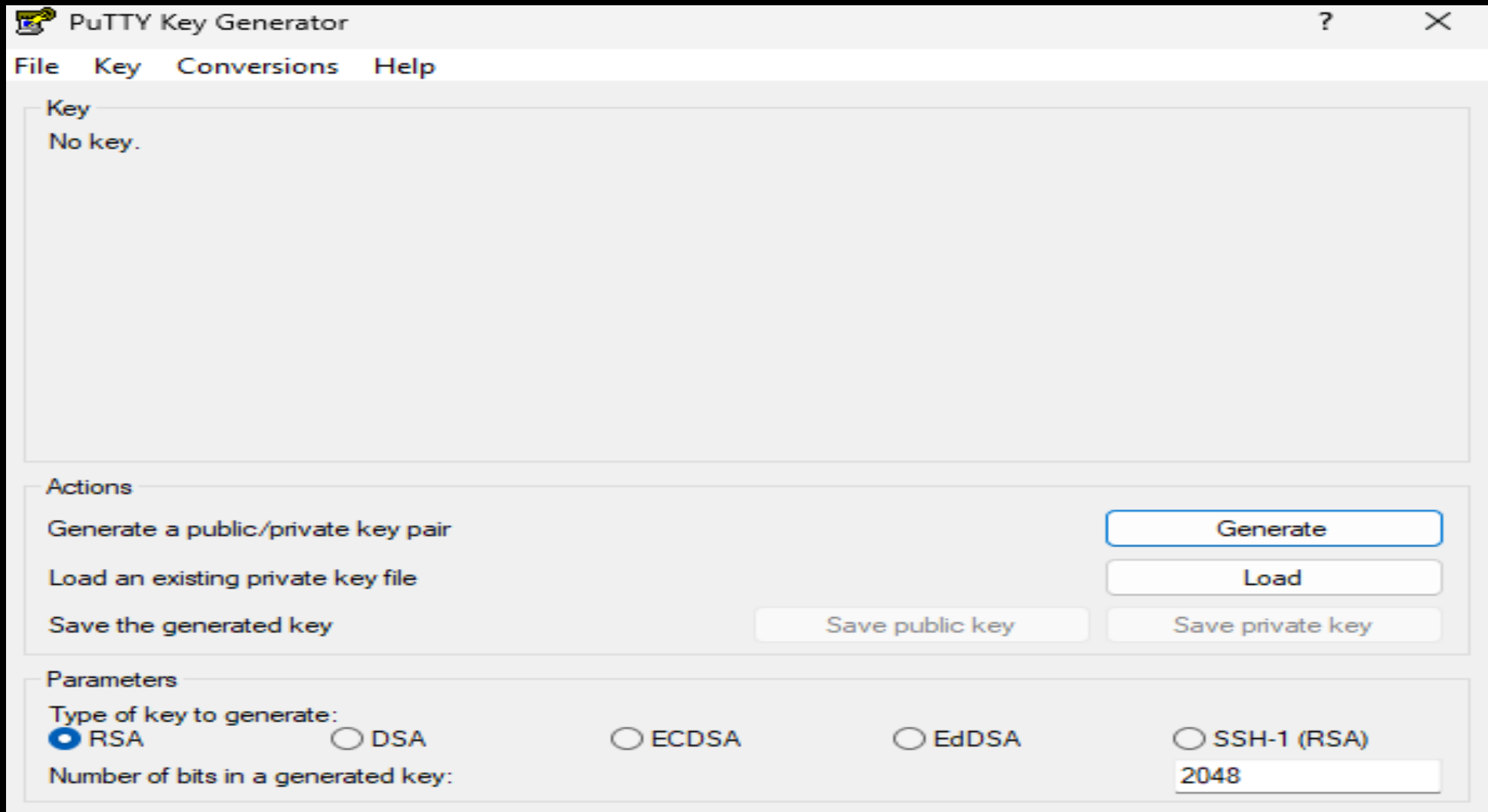
PuTTYgen is a tool used to generate and convert SSH key pairs. It allows you to convert AWS EC2 key pairs (.pem files) into PuTTY-compatible format (.ppk), which is required for securely connecting to your server via PuTTY. 🔑

Why We Use PuTTYgen? 🔑

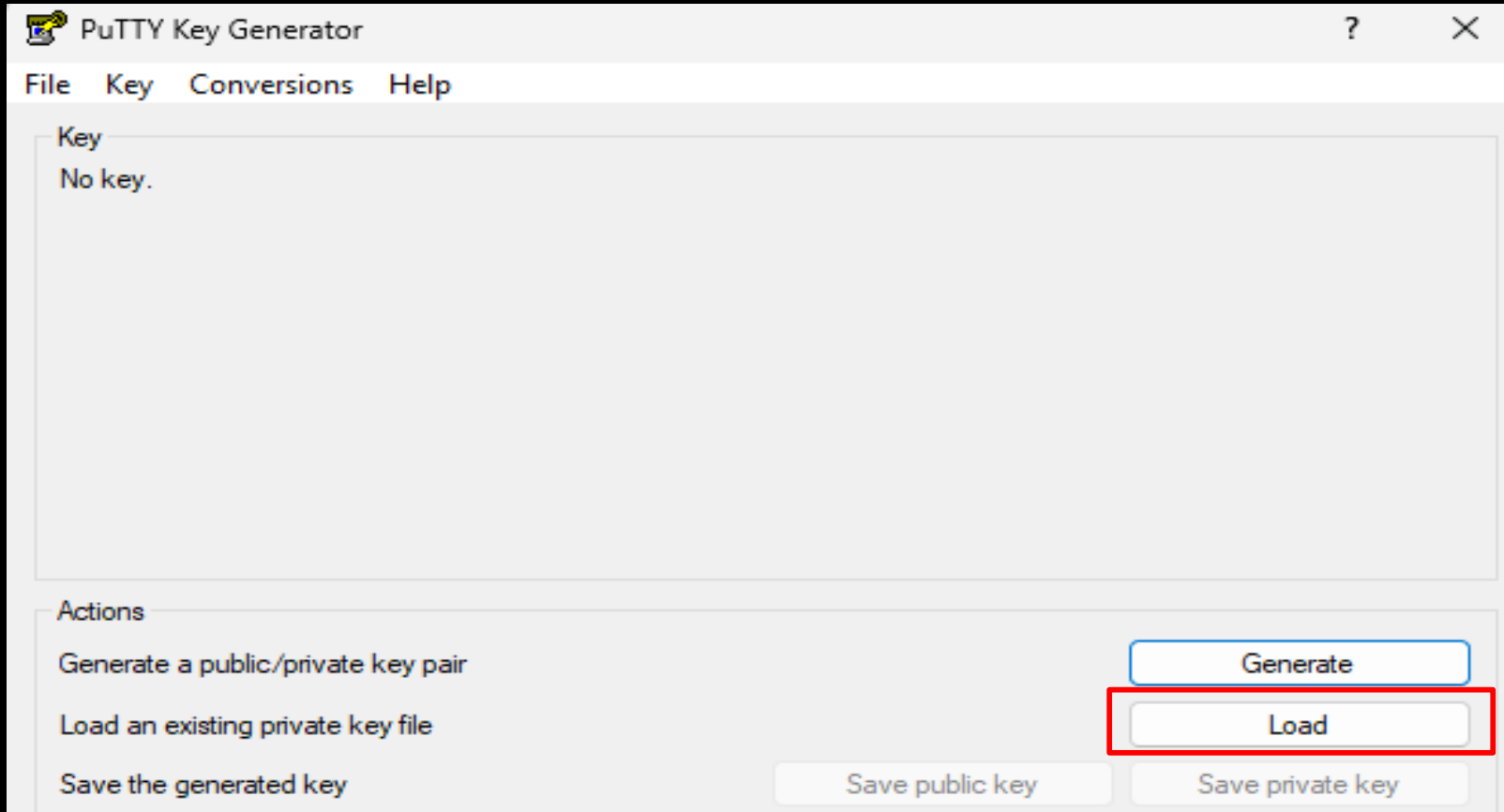
We use PuTTYgen to convert AWS EC2 key pairs (.pem files) into a PuTTY-compatible format (.ppk), allowing us to securely connect to a remote Linux server using PuTTY. Without this conversion, PuTTY cannot authenticate the connection. 💻

II. CONVERT YOUR .PEM FILE TO .PPK USING PUTTYGEN

1. Open PuTTYgen, your magical key converter! ✨



2. Click **Load** and select your freshly downloaded .pem.file. 🔒



 PuTTY Key Generator ? X

File Key Conversions Help

Key

Public key for pasting into OpenSSH authorized_keys file:

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCE73rSlo1OQu7qswnlgeBCBsbbbMRVqnYYbYW+re  
+7eJyINQPNjCrRaV01z6+zld41ZnVTsvmzsGSzL91b621fi0jknMBzmGUJtxo6cDHrxSgfH/86svhzD8zGyGVsRRPr  
7xdPQWWISpu5wSbH0t1865kxshDMOWjGxaC1amQO4CtxnaLRx72CgtsvZA3pt/1vvUbbOkeyyvYcJ8Fov  
+an8L0aQqTD083cL8SA1mQZMaUnCeh  
+23Fo0GuzcV4Qa6kUH0rWhCkdTLx2Plu721yUysjgWX/1gPtakOD7pa1HmkRaU6nXao9f4wyQlm73g6K/KxBub
```

Key fingerprint:

Key comment:

Key passphrase:

Confirm passphrase:

Actions

Generate a public/private key pair

Load an existing private key file

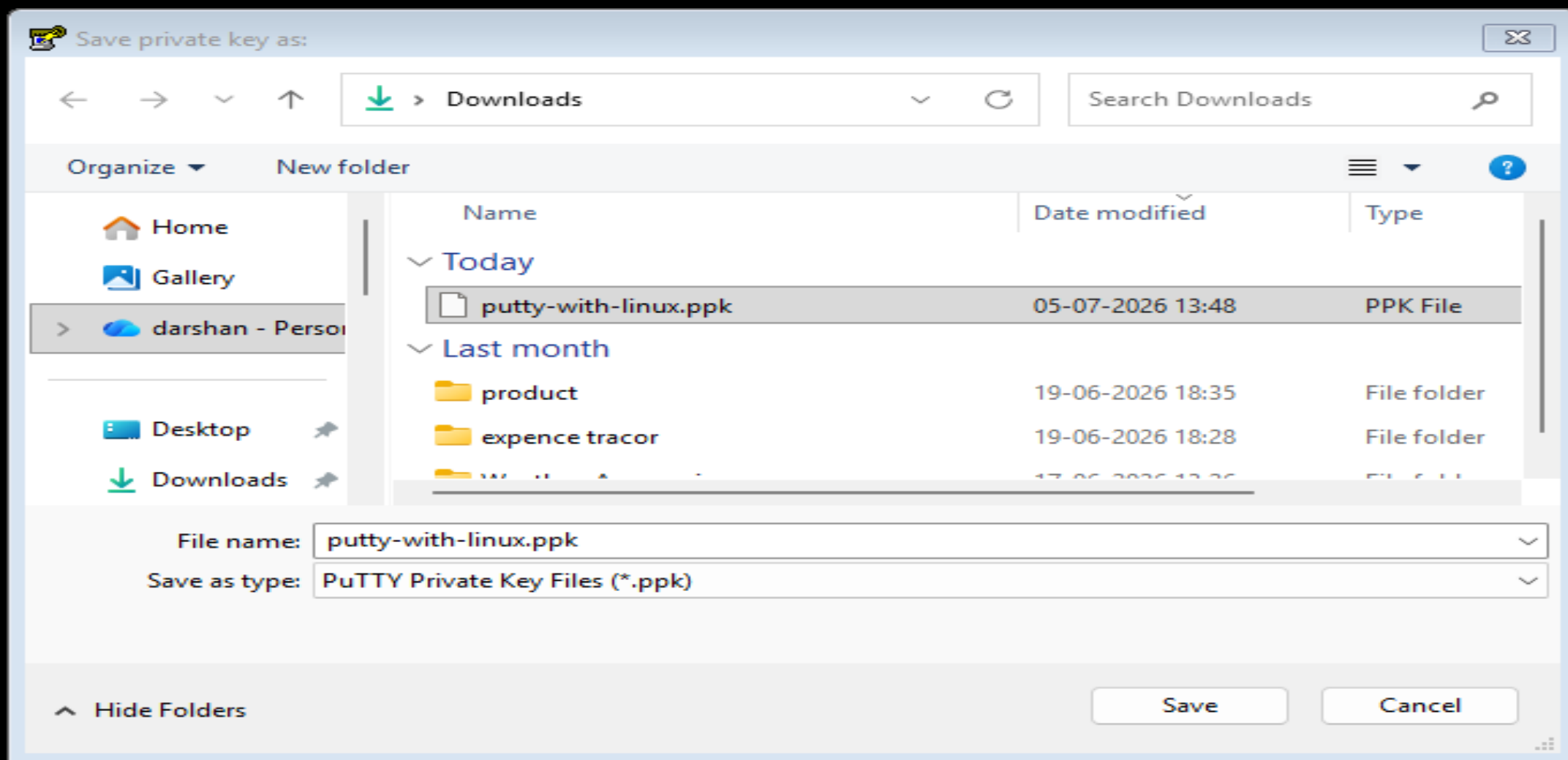
Save the generated key

Parameters

Type of key to generate:

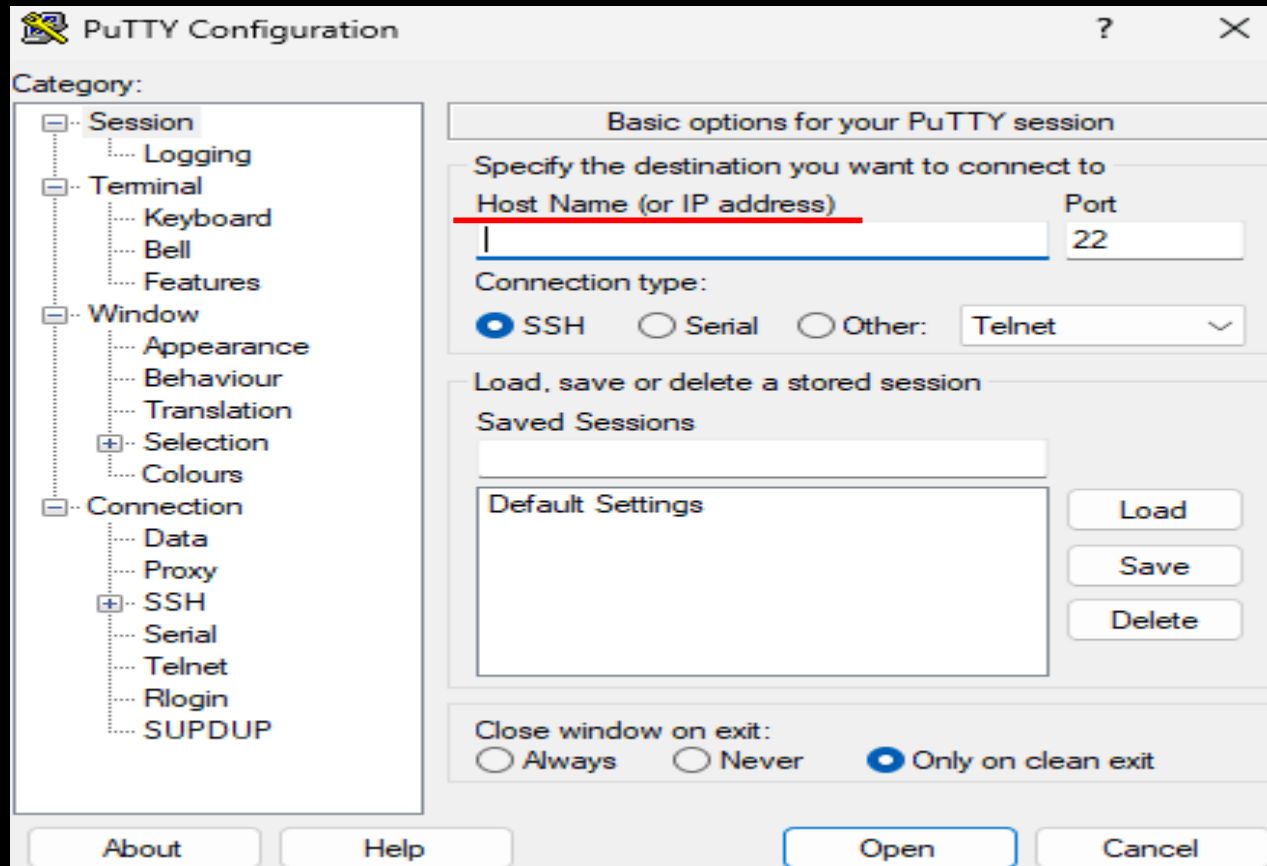
☒ RSA ☐ DSA ☐ ECDSA ☐ EdDSA ☐ SSH-1 (RSA)

Number of bits in a generated key:



III. CONNECT TO YOUR EC2 INSTANCE WITH PUTTY

1. Open **PuTTY** and get ready to teleport into your EC2 instance. 
2. In the **Host Name** field, enter the **Public IP Address** or **Public DNS** of your EC2 instance. 



S3

EC2

EC2

Instances

i-07c08217508a7e4af

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Instance summary for i-07c08217508a7e4af (linux-with-putty)

Updated less than a minute ago

Instance ID

i-07c08217508a7e4af

IPv6 address

-

Hostname type

IP name: ip-172-31-22-17.eu-north-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.61.190.156 [Public IP]

IAM role

-

IMDSv2

Required

Operator

Public IPv4 address

13.61.190.156 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-22-17.eu-north-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0ca3dd3c304fb2fcd

Subnet ID

subnet-03637d372f6e57462

Instance ARN

arn:aws:ec2:eu-north-1:543432151202:instance/i-07c08217508a7e4af

Private IPv4 addresses

172.31.22.17

Public DNS

ec2-13-61-190-156.eu-north-1.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

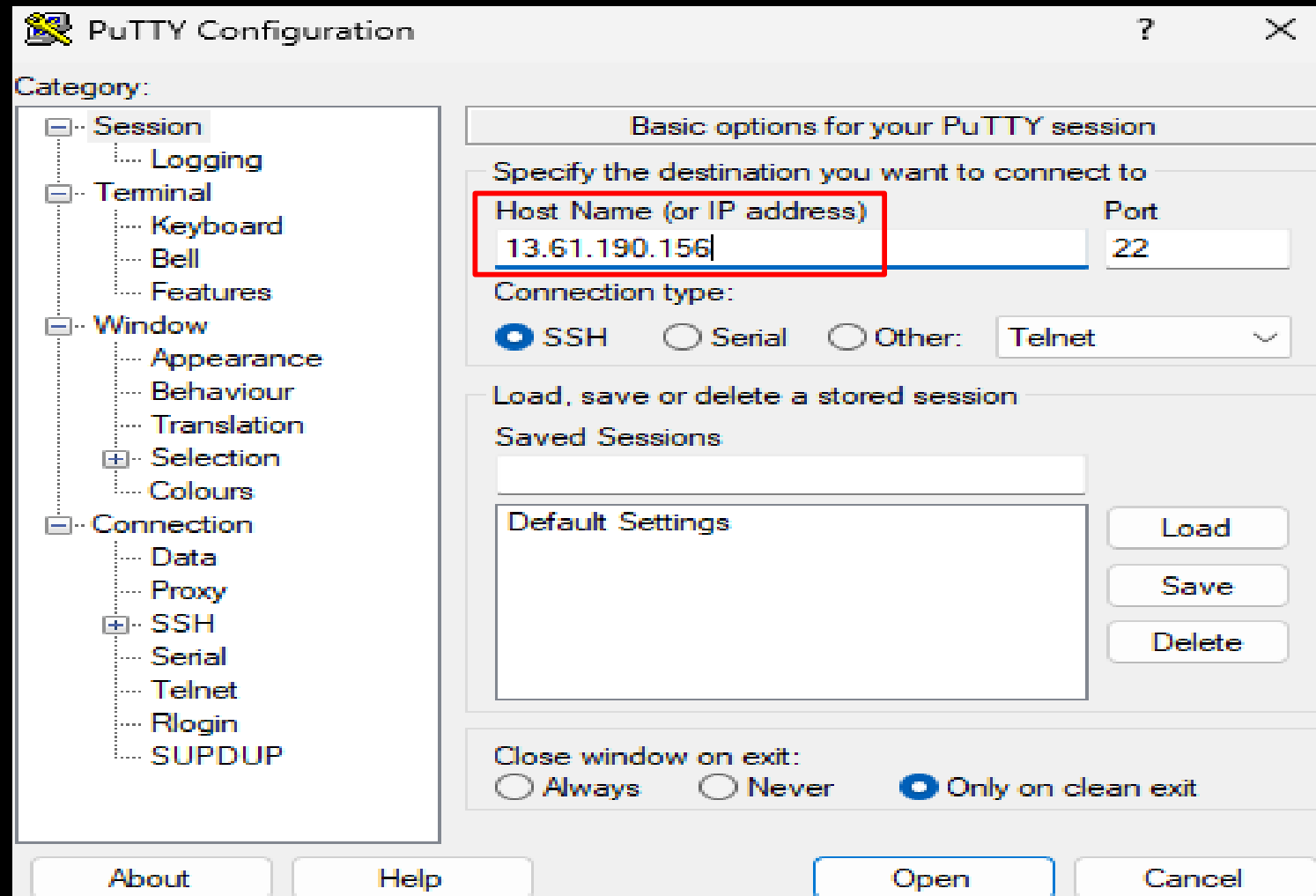
Managed

false

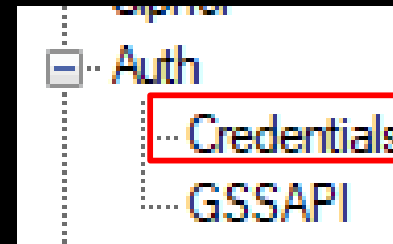
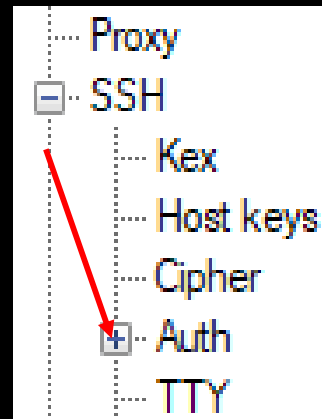
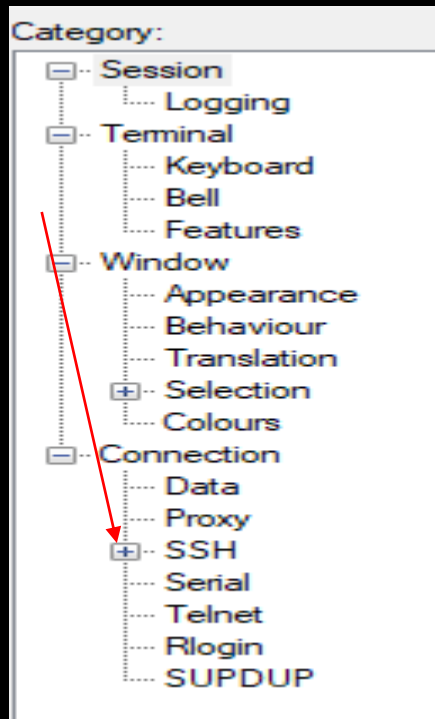
Connect

Instance state

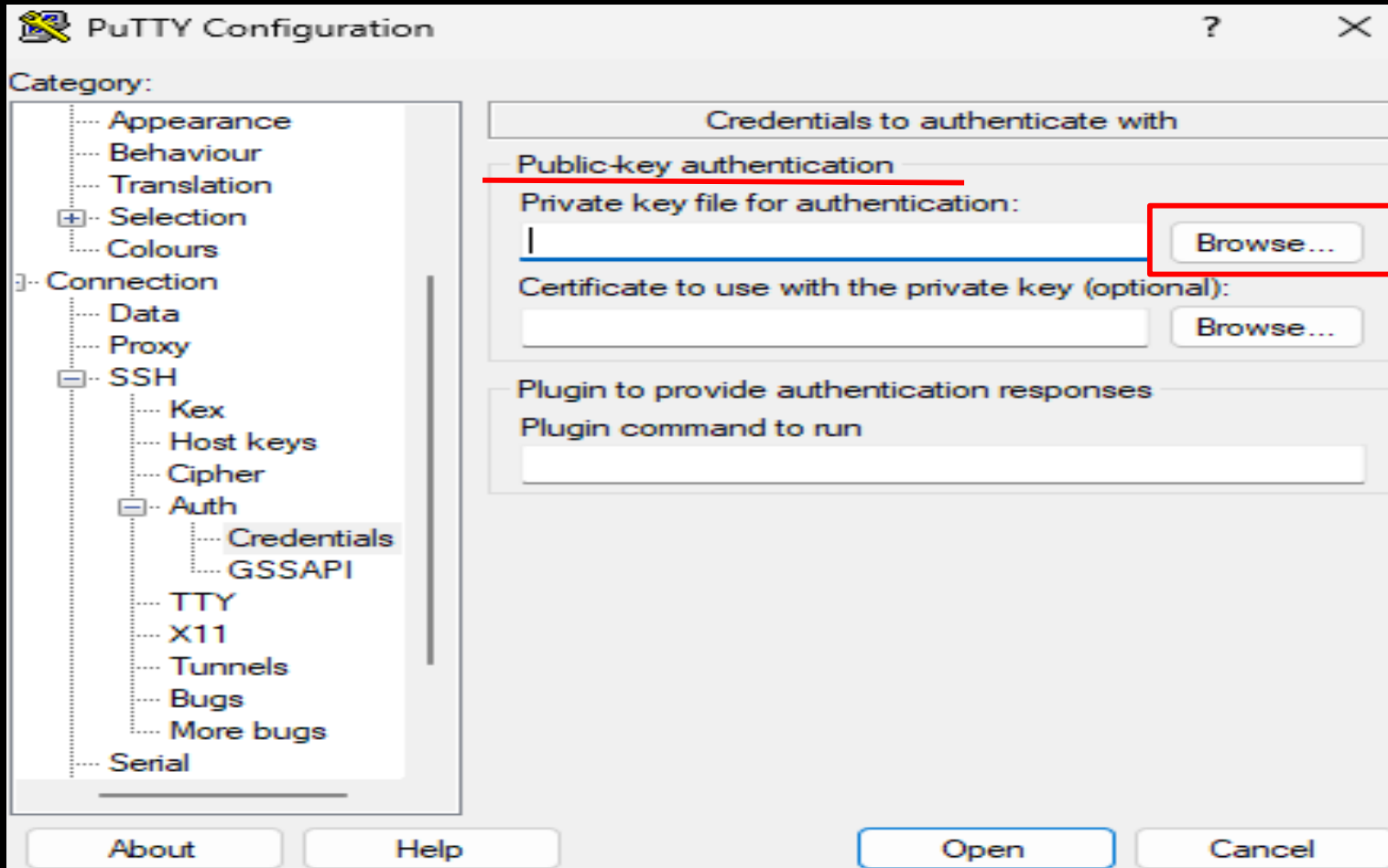
Actions

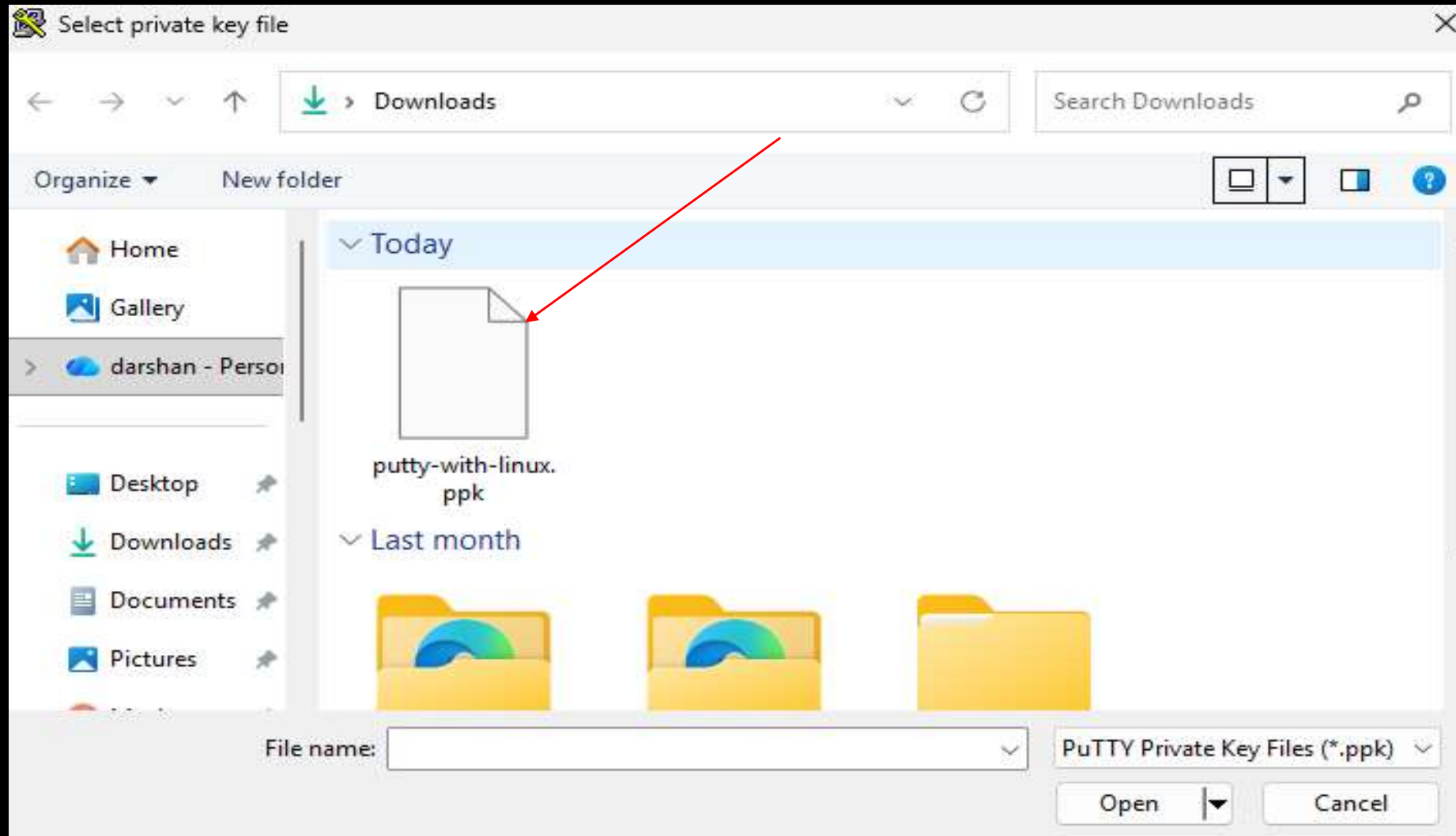


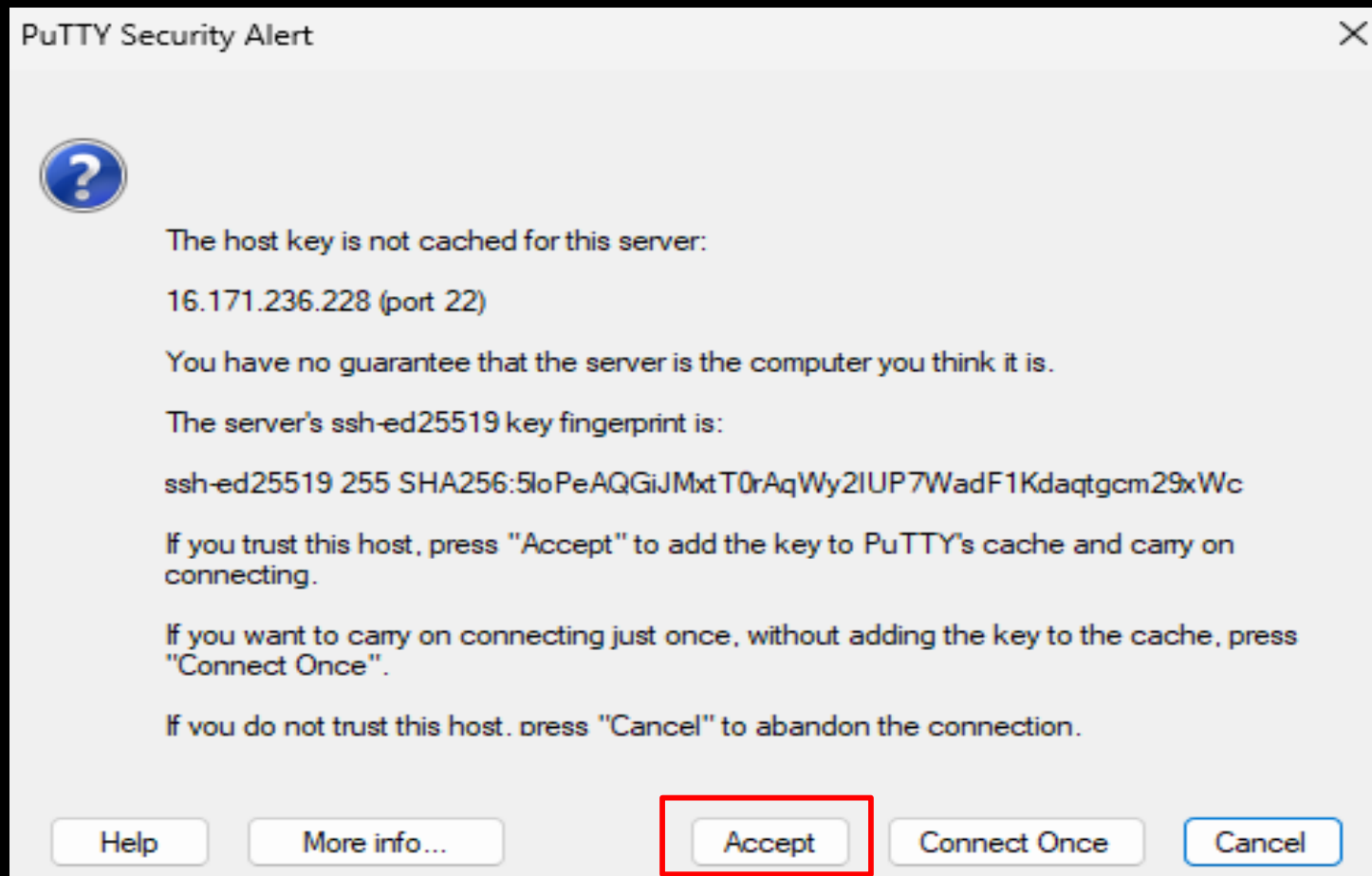
3. Go to **SSH > Auth** on the left menu.📁



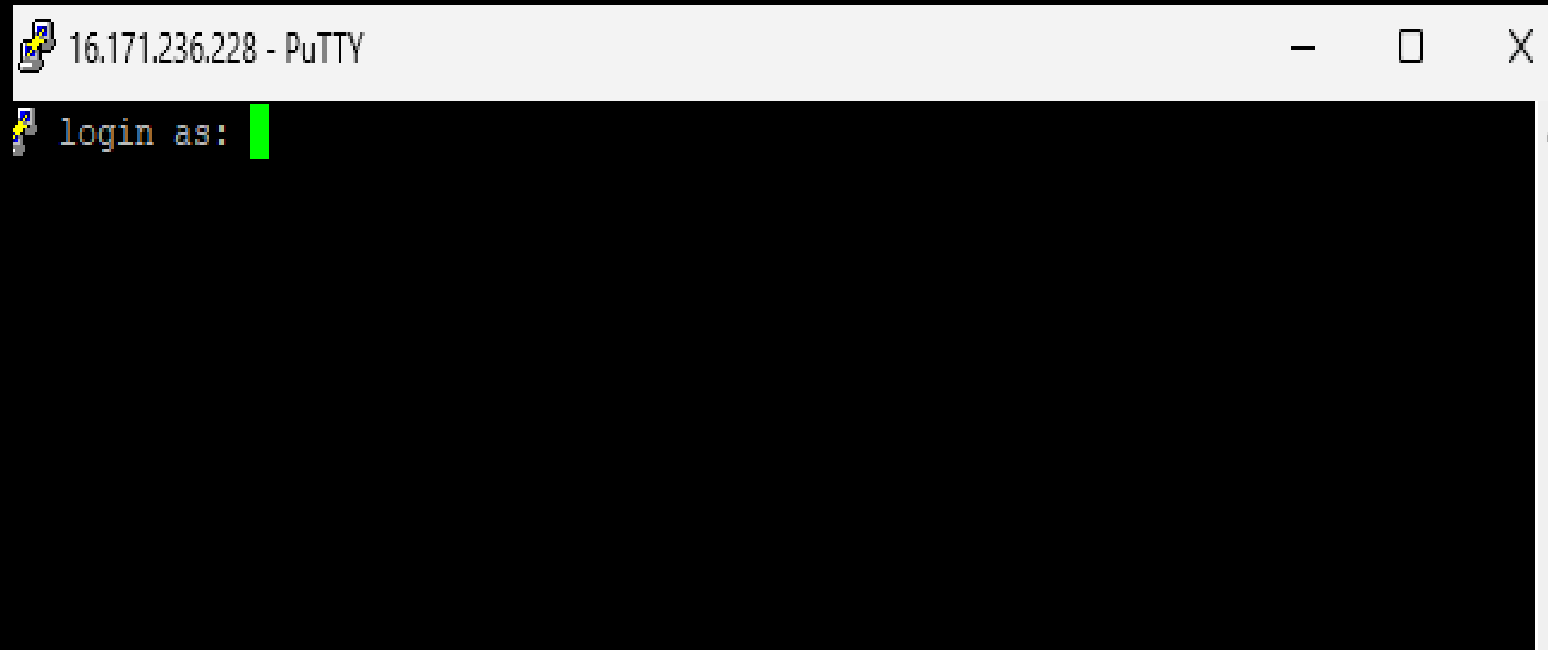
4 .Under Private key file for authentication, browse and select your .ppk file. 🔑







5. Hit **Open** and voilà, you're almost there!



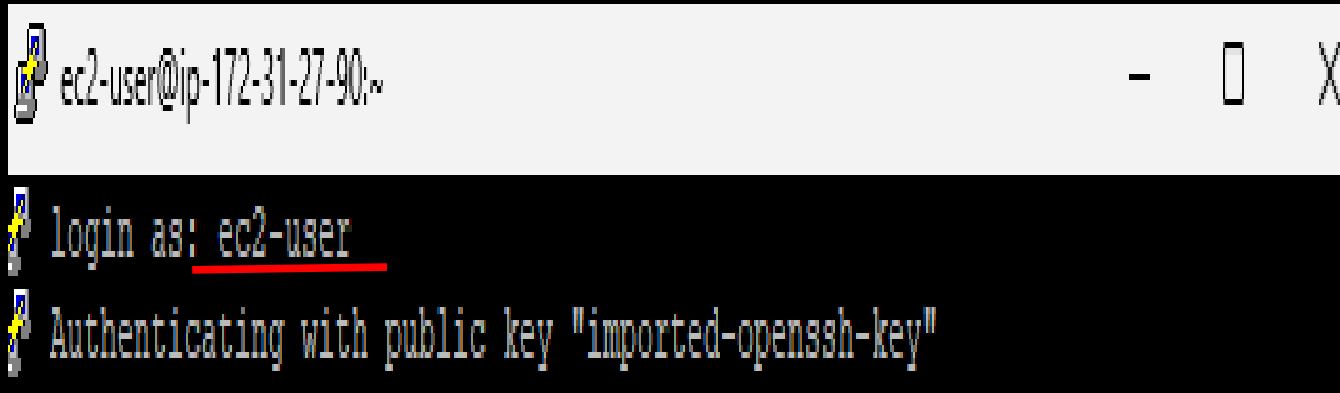
IV. LOGIN LIKE A PRO

1. Depending on your AMI, use the **default username** to log in: -

Amazon Linux: ec2-user

Ubuntu: ubuntu

CentOS: centos



```
ec2-user@ip-172-31-27-90:~  
login as: ec2-user  
Authenticating with public key "imported-openssh-key"
```

2. You're officially inside your shiny new Linux machine! 🖥️✨

```
ec2-user@ip-172-31-27-90:~  
login as: ec2-user  
Authenticating with public key "imported-openssh-key"  
  
#_  
~\#### Amazon Linux 2023  
~~\_#####\  
~~\_###|  
~~\_#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~V~' '->  
~~~  
~~.-.-/  
~/m/'
```

[ec2-user@ip-172-31-27-90 ~]\$ █



Amazon
EC2

THANK YOU

